

ams-dist: **A Scalable Distributed Hybrid Solver for Hard Combinatorial Problems**

Maxim Zhulin, Piyush Jha, Zhengyu Li, Richard Vuduc, Vijay Ganesh

Modern Distributed Cube & Conquer Solvers?

- Portfolio Solvers have recently dominated SAT Competitions

Parallel Track

	Solver	Authors	PAR-2	Solved
1	MallobSat	Dominik Schreiber, Niccolo Rigi-Luperti, and Armin Biere	394.56	337
2	PL-PRS-Kissat	Mazigh Saoudi, Souheib Baarir, Julien Sopena, Thibault Lejembale, and Sabrine Saouli	406.87	334
3	PRS-SC25-SBVA	Yuhang Qian, Zhihan Chen, Xindi Zhang, and Shaowei Cai	457.19	331

Results from 2025 SAT Competition ([link](#)) [1]

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- Cube & Conquer solvers can still be better than Portfolio
 - Solving large problems with domain knowledge
 - Solving ALLSAT via blocking clauses

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- **Portfolio**
 - When # cubes < # cores, use portfolio solving to simplify cubes
 - Improve cubing variable quality
 - Solve easy cubes faster

AlphaMapleSAT-CnC

- AlphaMapleSAT-CnC has a lot of idle core time

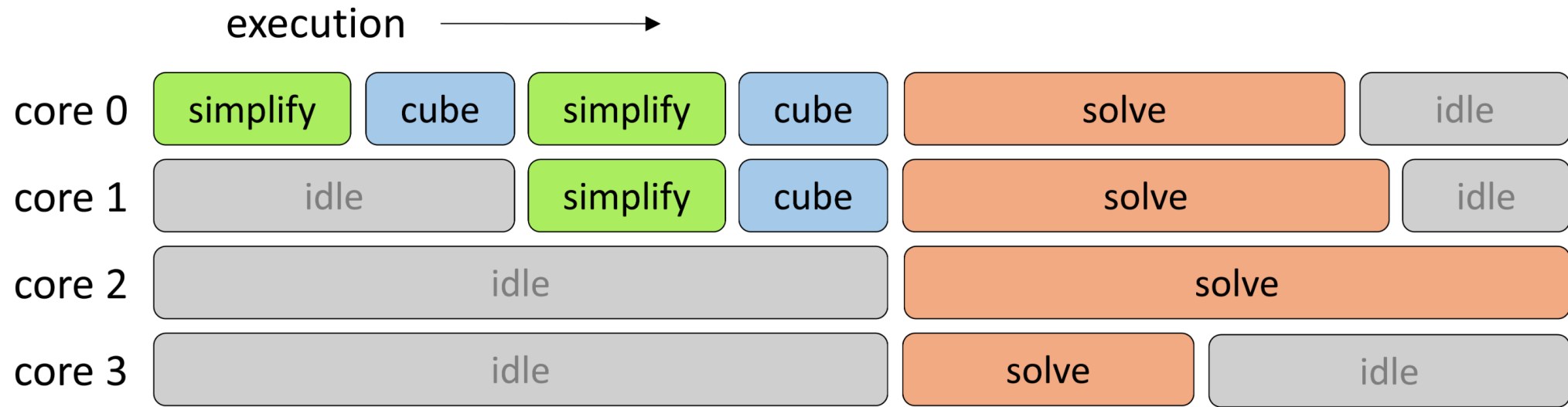


Figure 1: An example execution of *AlphaMapleSAT-CnC* on 4 cores

ams-dist: a hybrid approach

- MPI distributed implementation
- Replaces single-core simplification with portfolio simplification
- Distributed cubing step
- Work-stealing based on a progress metric during the conquering phase

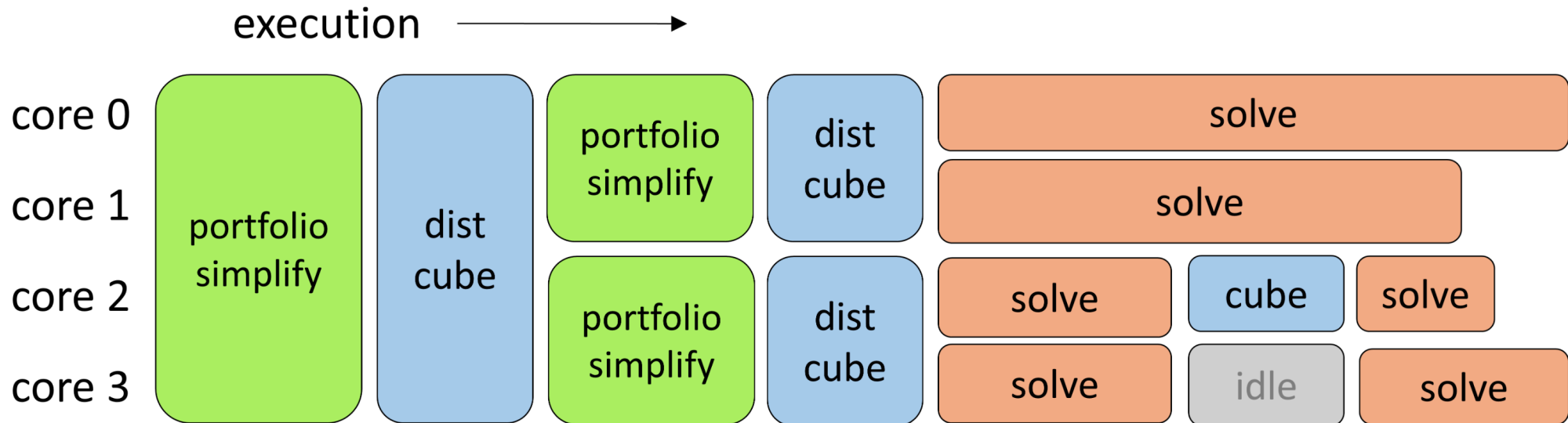


Figure 2: An example execution of *ams-dist* on 4 cores

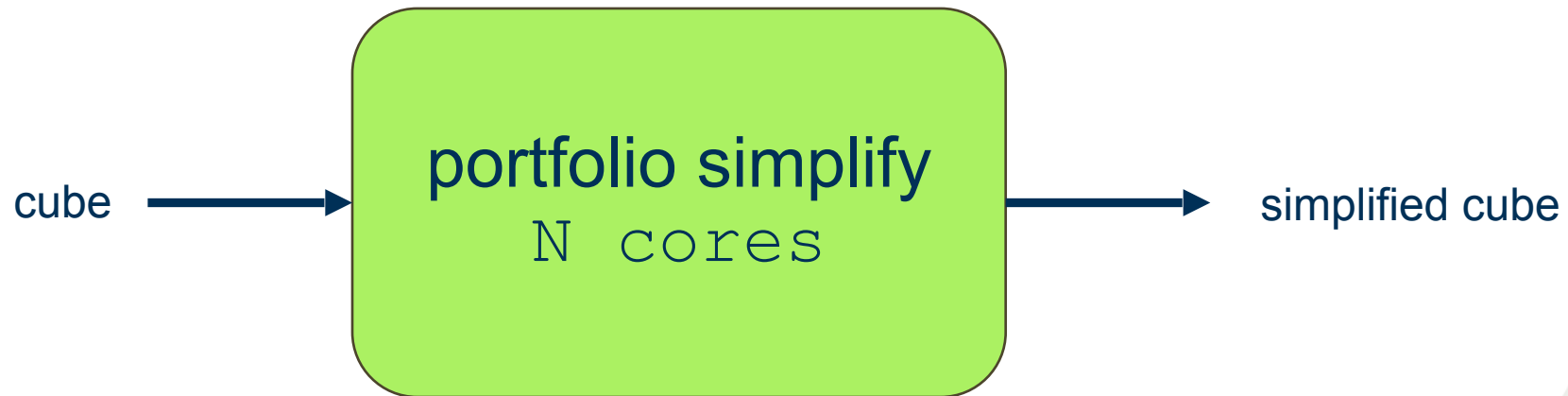
ams-dist: Portfolio Simplification

- **AlphaMapleSAT-CnC** simplifies a cube for 10,000 conflicts before cubing further



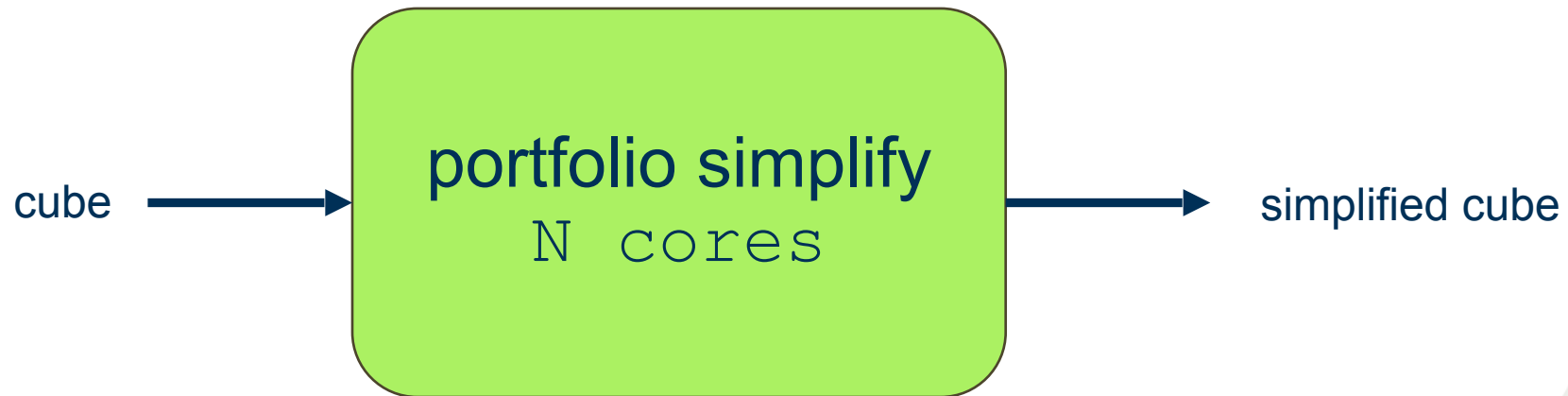
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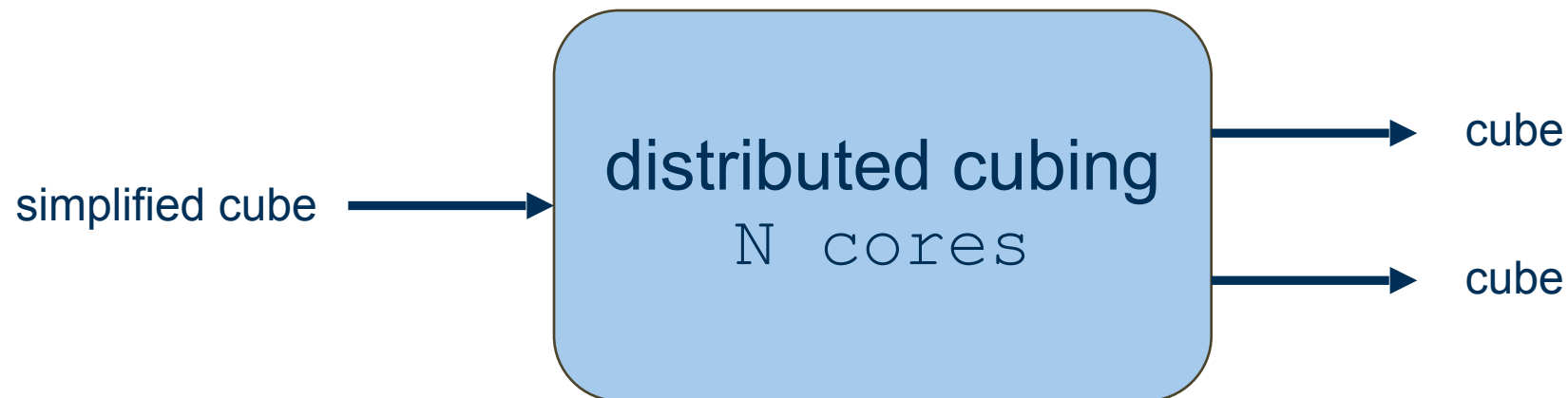
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- Ideally leads to better splitting variable choices



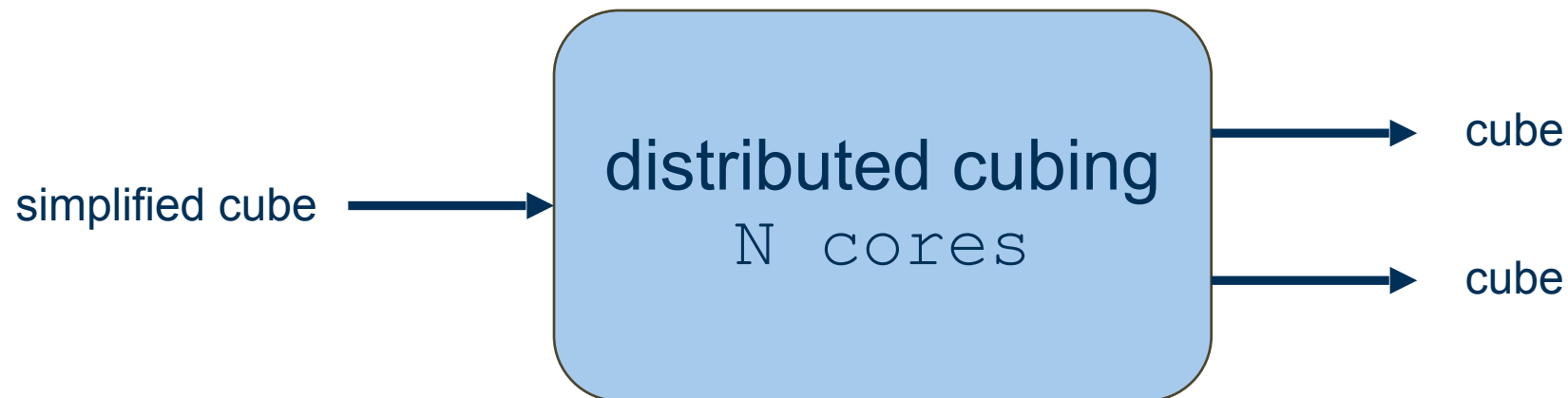
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- **AlphaMapleSAT-CnC** uses a Monte-Carlo Tree Search with a novel evaluation function to pick the best splitting variable



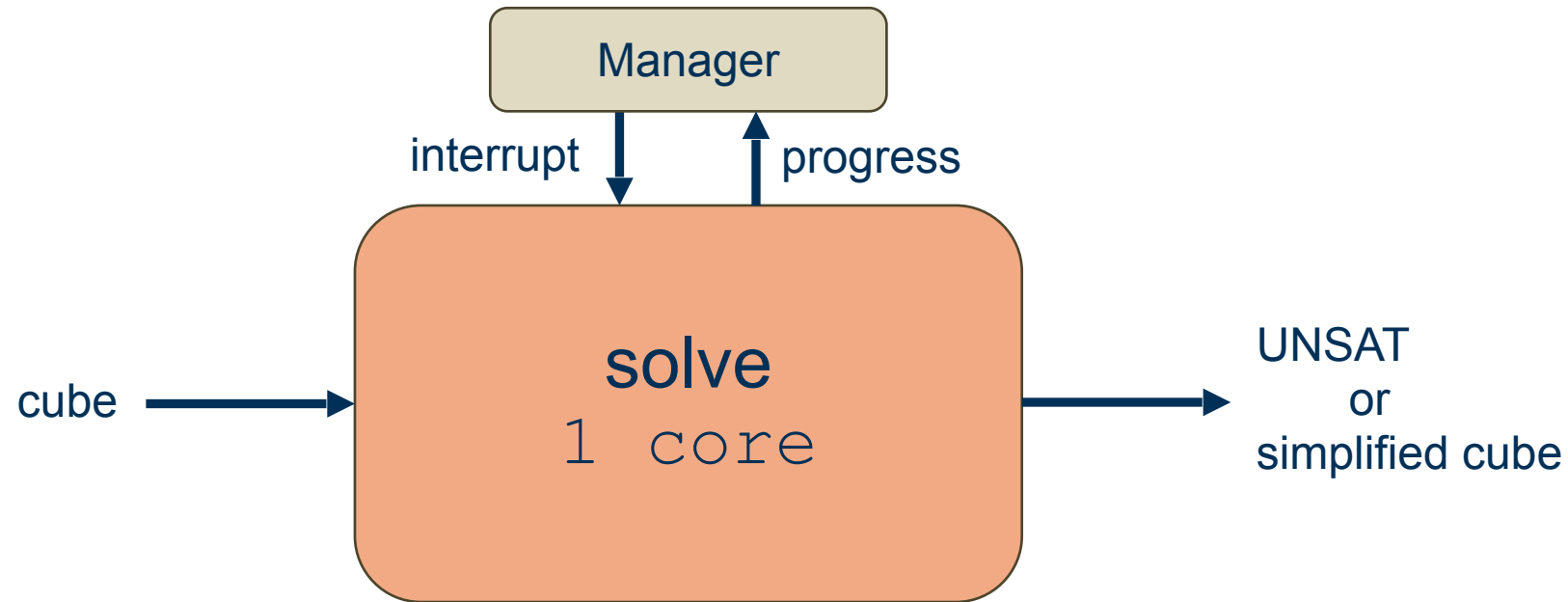
ams-dist: Distributed Cubing

- **AlphaMapleSAT-CnC** uses a Monte-Carlo Tree Search with a novel evaluation function to pick the best splitting variable
- **ams-dist** takes the evaluation function from AlphaMapleSAT-CnC but evaluate ALL splitting variables by **distributing the computation** across the involved cores



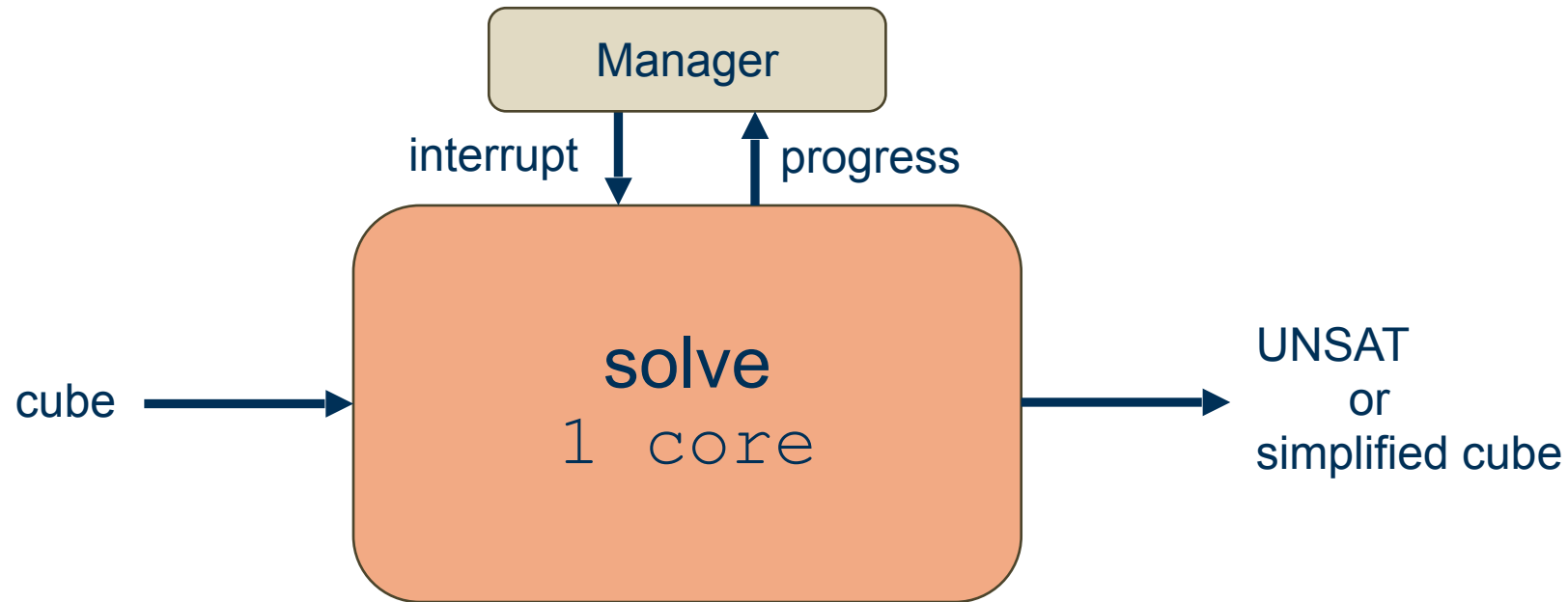
ams-dist: Work Stealing

- During solving, all cores report a progress heuristic based on the number of irredundant clauses



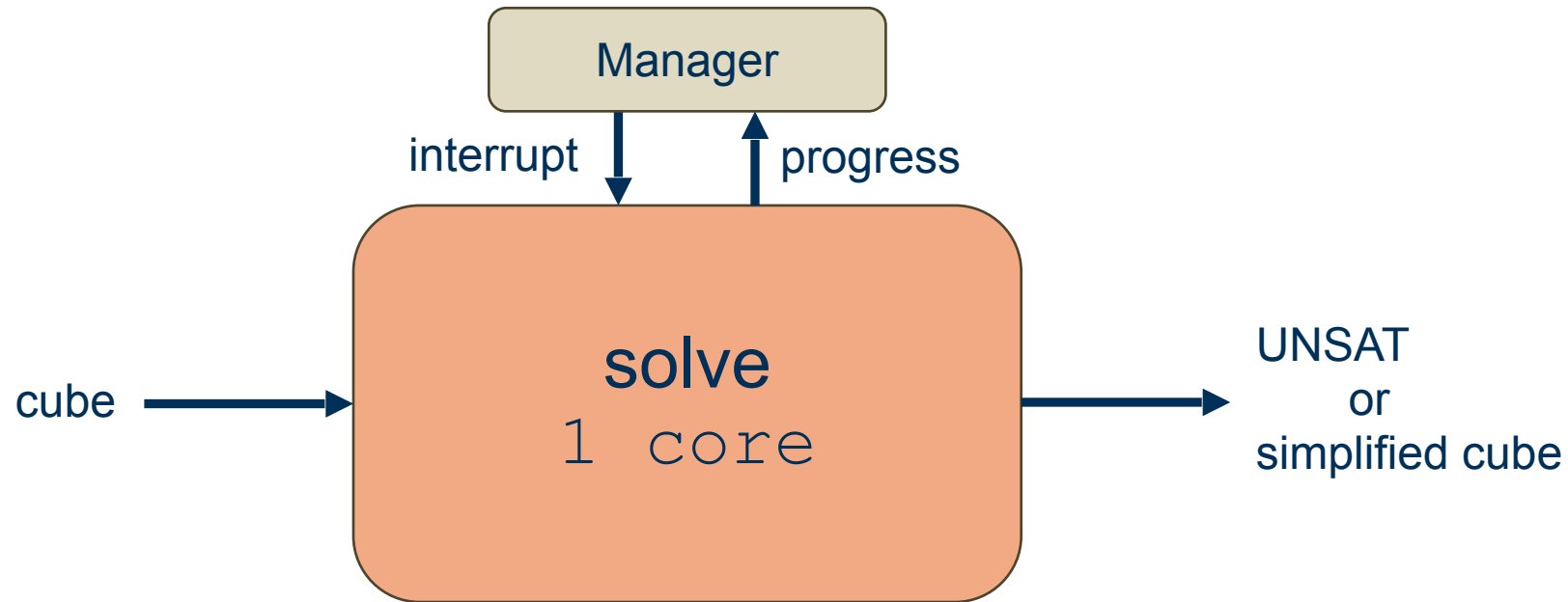
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- When a core solves its cube and requires work, the central manager **interrupts** the core working on the “hardest” problem and splits it further



Benchmark: Minimum Kochen-Specker Problem

- Kochen-Specker (KS) Theorem in Physics
 - Prohibits certain hidden-variable theorems
 - Proven using existence of a Kochen-Specker set

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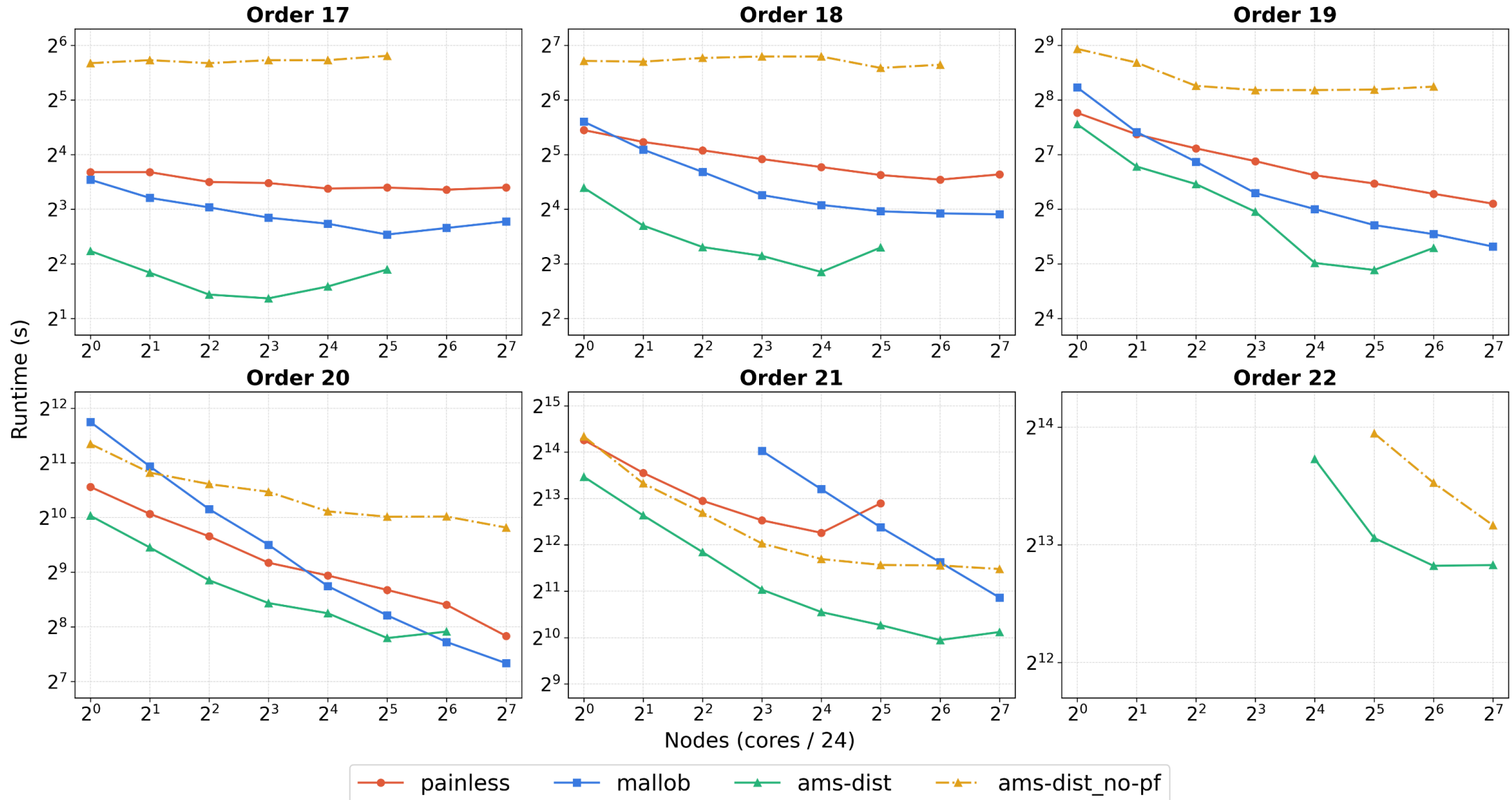
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- Lower Bounds
 - 24 – Established by a SAT+CAS Cube & Conquer Approach (Li, 2024) [6]
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- Difficulty grows very quickly with KS set size (order)
 - Proving lower bound 24 took 12,000 CPU hours

Performance and Scaling



Future Work

- Improving portfolio simplification step
 - Using SOTA portfolio techniques
 - MPI + threads

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- Dynamically allocate cores to cubes during portfolio solving based on difficulty

Thank you!

- Implementation
 - <https://github.com/map0te/ams-dist>

References

- [1] C. Codel, K. Fazekas, M. J. H. Heule, M. Iser (Eds.), Proceedings of SAT Competition 2025: Solver and Benchmark Descriptions, 2025. URL: <https://satcompetition.github.io/2025/>.
- [2] D. Schreiber, P. Sanders, Scalable SAT solving in the cloud, in: Theory and Applications of Satisfiability Testing – SAT 2021, volume 12831 of Lecture Notes in Computer Science, Springer, 2021, pp. 518–534.
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- [6] Z. Li, C. Bright, V. Ganesh, A SAT solver + computer algebra attack on the minimum Kochen–Specker problem, in: Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI-24), 2024, pp. 1898–1906.
- [7] A. Peres, Quantum Theory: Concepts and Methods, Kluwer Academic Publishers, Dordrecht, 1993; the 31-vector Kochen–Specker system is attributed to J. H. Conway and S. Kochen.